HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2024)

Student No.: 04 Name: _____ Marks: / 70

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Denote the origin by O.

(a) A and B are points lying on the positive x-axis such that the x-coordinate of A is greater than the x-coordinate of B. A vertical line which passes through B cuts the straight line $y = mx$ at the point C such that $AB = BC$, where m is a positive constant. Let D be a point such that $ABCD$ is a square. Express the slope of OD in terms of m. (3 marks)

(b) The coordinates of the points M and N are $(6, 5)$ and $(10, 0)$ respectively. Let P and Q be points lying on OM and MN respectively while R and S be points lying on the x-axis. If the quadrilateral $PQRS$ is a square, find the x-coordinate of P. (4 marks)

(7 marks) (DSE-2024 PI Q12)

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2. Let $F(x) = (6x^2 + x + p)(qx^2 + rx - 10)$, where p, q and r are constants. The constant term of $F(x)$ is 40.

(a) Write down the value of p. (1 mark)

(b) When $F(x)$ is divided by $x + 1$, the remainder is -12. It is given that $x - 2$ is a factor of $F(x)$.

(i) Find q and r.

(ii) How many irrational roots does the equation $F(x) = 0$ have? Explain your answer. (7 marks)

(8 marks) (DSE-2024 PI Q14)

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3. It is given that $\log_9 y$ is a linear function of $\log_3 x$. Denote the graph of the linear function by L. The slope of L is 4 and L passes through the point $(5, 22)$. Express y in terms of x. (3 marks) (DSE-2024 PI Q15)

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4. In a bag, there are 16 red cups and 4 white cups. If 5 cups are randomly drawn from the bag at the same time, find

(a) the probability that exactly 1 white cup is drawn; (2 marks)

(b) the probability that at most 3 red cups are drawn. (2 marks)

(4 marks) (DSE-2024 PI Q16)

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5. The coordinates of the points Q and R are $(10, -1)$ and $(-4, -9)$ respectively.

(a) Let P be a moving point in the rectangular coordinate plane such that $PQ = PR$. Denote the locus of P by Γ.

(i) Describe the geometric relationship between Γ and QR.

(ii) Find the equation of Γ. (3 marks)

(b) Let C be the circle which passes through Q, R and the point $(4, 3)$.

(i) Find the equation of C.

(ii) The coordinates of the point U are $(10, 4)$. It is found that U lies outside C. UV and UW are the tangents to C at the points V and W respectively. Is the area of the circumcircle of $\triangle UVW$ greater than 100? Explain your answer. (5 marks)

(8 marks) (DSE-2024 PI Q17)

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6. (a) $PQRS$ is a thin quadrilateral metal sheet, where $PQ = 12\,\text{cm}$, $PS = 10\,\text{cm}$, $QR = 13\,\text{cm}$, $\angle QPS = 82^\circ$ and $\angle QRS = 65^\circ$. Find

(i) the length of QS,

(ii) $\angle RQS$. (4 marks)

(b) The metal sheet $PQRS$ described in (a) is now folded along QS (see Figure 2). It is given that the angle between the plane PQS and the plane QRS is 80°.

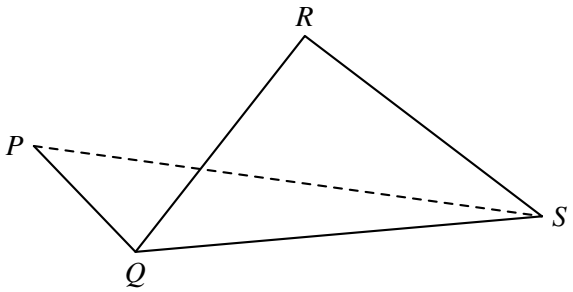


Figure 2

(i) Find the shortest distance from R to the plane PQS.

(ii) Let X be any point lying on the plane QRS. Someone claims that the distance between P and X exceeds $8\,\text{cm}$. Is the claim correct? Explain your answer. (4 marks)

(8 marks) (DSE-2024 PI Q18)

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7. Let $f(x) = 2x^2 + 4mx + 8x + 2m^2 + 8m + n$, where m and n are real constants such that $mn < 0$. Denote the vertex of the graph of $y = f(x)$ by P.

(a) Using the method of completing the square, express the coordinates of P in terms of m and n. (2 marks)

(b) Describe the geometric meaning represented by transforming $f(x)$ to $f\left(\frac{x}{5}\right) + 7$. (2 marks)

(c) Denote the vertex of the graph of $y = f\left(\frac{x}{5}\right) + 7$ by Q. Let (a_1, b_1) and (a_2, b_2) be the coordinates of P and Q respectively. It is given that a_1, $1 + n$, a_2 is an arithmetic sequence and b_1, $4 - m$, b_2 is a geometric sequence.

(i) Find the coordinates of P and Q.

(ii) The coordinates of the points R and S are $(3t + 27, t)$ and $(3t + 3, 2t - 3)$ respectively, where t is a real number. Is it possible that $PQRS$ is a rhombus? Explain your answer. (8 marks)

(12 marks) (DSE-2024 PI Q19)

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8. Let $f(x) = x^3 + kx^2 + 5x + 10$, where k is a constant. If $f(x)$ is divisible by $x + k$, find the remainder when $f(x)$ is divided by $x + 1$.

A. -2

B. 2

C. 6

D. 18

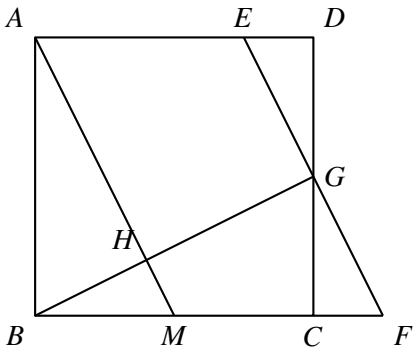
(2 marks) (DSE-2024 PII Q09)

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9. In the figure, $ABCD$ is a square. Let M be the mid-point of BC. E is a point lying on AD such that $AE : ED = 3 : 1$. F is a point lying on BC produced such that $EF//AM$. CD and EF intersect at the point G while AM and BG intersect at the point H. If the area of $\triangle BMH$ is $4\,\text{cm}^2$, then the area of the trapezium $AEGH$ is



A. $12\,\text{cm}^2$.

B. $33\,\text{cm}^2$.

C. $39\,\text{cm}^2$.

D. $45\,\text{cm}^2$.

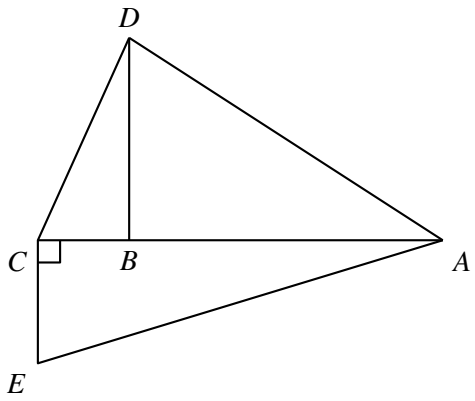
(2 marks) (DSE-2024 PII Q17)

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10. In the figure, ABC is a straight line. It is given that $AD = 37\,\text{cm}$, $BC = 5\,\text{cm}$, $BD = 12\,\text{cm}$, $CD = 13\,\text{cm}$ and $CE = 9\,\text{cm}$. If $\angle ACE = 90°$, find the perimeter of the quadrilateral $ADCE$.



A. 76 cm

B. 90 cm

C. 100 cm

D. 180 cm

(2 marks) (DSE-2024 PII Q18)

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11. If the sum of the interior angles of a regular polygon is 900°, which of the following is/are true?

I. The number of diagonals of the polygon is 7.

II. The number of folds of rotational symmetry of the polygon is 7.

III. The number of axes of reflectional symmetry of the polygon is 7.

A. I only

B. II only

C. I and III only

D. II and III only

(2 marks) (DSE-2024 PII Q20)

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12. Denote the circle $3x^2 + 3y^2 - 6x + 12y - 4 = 0$ by C. Which of the following are true?

I. The origin lies inside C.

II. The circumference of C is less than 16.

III. The perpendicular distance from the centre of C to the x-axis is 2.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q27)

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13. Consider the following positive integers:

$$5 \quad 5 \quad 5 \quad 6 \quad 9 \quad 9 \quad 11 \quad 13 \quad m \quad n$$

Let p, q and r be the standard deviation, the mode and the median of the above positive integers respectively. If the mean of the above positive integers is 7, which of the following must be true?

I. $p > 3$

II. $q = 5$

III. $r < 7$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q30)

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14. $\mathrm{AF000000000BC}_{16} =$

A. $175 \times 16^{11} + 188.$

B. $192 \times 16^{11} + 205.$

C. $175 \times 16^{12} + 188.$

D. $192 \times 16^{12} + 205.$

(2 marks) (DSE-2024 PII Q32)

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15. Consider the following system of inequalities:

$$\begin{cases} x - 2y \leq 1 \\ x + 4y \leq 13 \\ 2x - y \geq -1 \end{cases}$$

Let R be the region which represents the solution of the above system of inequalities. Find the constant c such that the least value of $5x - 2y + c$ is 22, where (x, y) is a point lying in R.

A. 1

B. 23

C. 25

D. 29

(2 marks) (DSE-2024 PII Q37)

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16. For $0° < \theta \leq 360°$, how many roots does the equation $4\cos^2\theta - 3\cos\theta - 1 = 0$ have?

A. 3

B. 4

C. 5

D. 6

(2 marks) (DSE-2024 PII Q39)

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17. Let G, H, I and J be the centroid, the orthocentre, the in-centre and the circumcentre of $\triangle PQR$ respectively. If $\angle PQR = \angle PRQ = 22°$, which of the following are true?

I. G lies inside $\triangle PQR$.

II. H lies outside $\triangle PQR$.

III. I, J and Q are collinear.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q41)

I can finish _____ questions

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